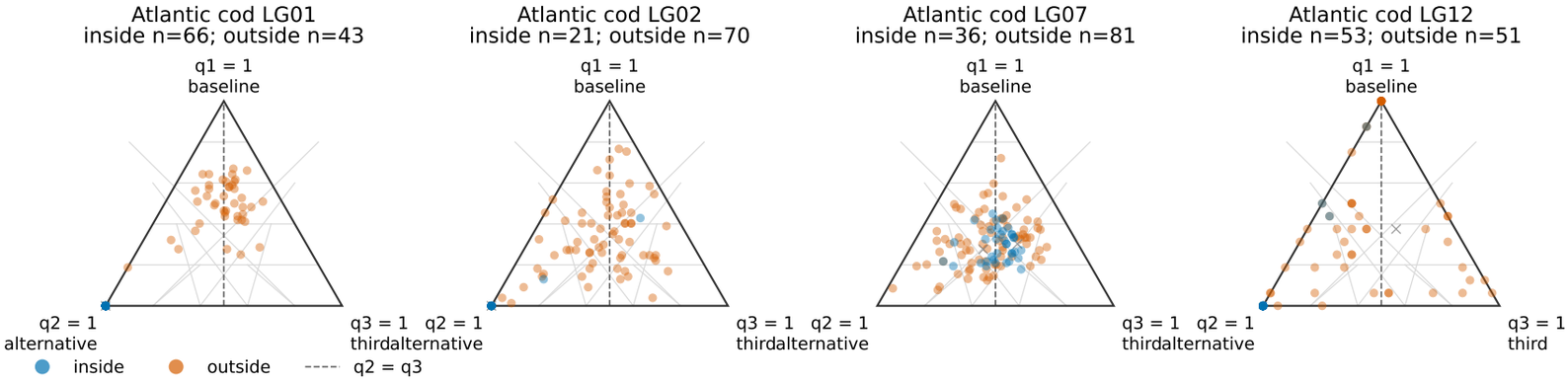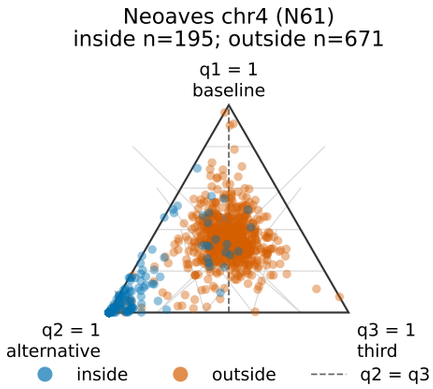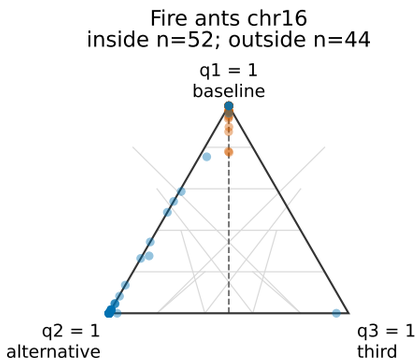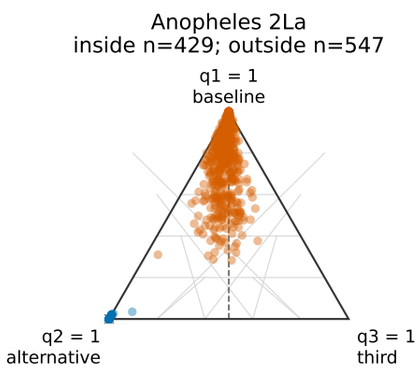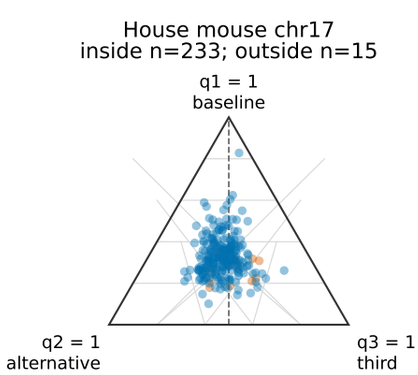Empirical quartet-support simplexes inside and outside rearranged regions